

# DATABASE APPLICATIONS (240)

## NATIONAL – 2019

**Certification Exam:** MOS Access 2016 \_\_\_\_\_ (100 points)

**Production:**

Printout 1a: Person Table \_\_\_\_\_ (37 points)

Printout 1b: Relationship Report \_\_\_\_\_ (35 points)

Printout 2a: Add Close Form Button \_\_\_\_\_ (30 points)

Printout 2b: Conventions Form \_\_\_\_\_ (85 points)

Printout 3a: Add Command Buttons \_\_\_\_\_ (45 points)

Printout 3b: Conventions Summary Form \_\_\_\_\_ (60 points)

Printout 4: Convention Query \_\_\_\_\_ (45 points)

Printout 5: Convention Revenue Potential Query \_\_\_\_\_ (60 points)

Printout 6: Convention Revenue Report \_\_\_\_\_ (52 points)

**TOTAL POINTS** \_\_\_\_\_ **(549 points)**

**Failure to adhere to any of the following rules will result in disqualification:**

1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

No more than ten (10) minutes orientation

No more than 90 minutes testing time

No more than ten (10) minutes wrap-up

Property of Business Professionals of America.  
May be reproduced only for use in the Business Professionals of America  
Workplace Skills Assessment Program competition.

## **General Instructions**

1. Ensure this test booklet contains Jobs 1-6.
2. Your name, initials, and school should *not* appear on any work you submit. Instead, key your contestant number everywhere xx-xxxx-xxxx appears.
3. If you finish before the end of the allotted testing time, notify the proctor. Time may be a factor in determining a winner when there is a tie score.
4. When turning in your completed work, order the printed pages by job number, and place your scoring sheet on top of all of your jobs.

## DATABASE APPLICATIONS SCORING SHEET

| <i>Unless indicated otherwise, student receives all points or none.</i>     | <b>Points Possible</b> | <b>Points Earned</b> |
|-----------------------------------------------------------------------------|------------------------|----------------------|
| <b>Printout 1a – Person Table (37 points maximum)</b>                       |                        |                      |
| Report printed portrait and all columns are visible – Print both pages      | 5                      |                      |
| Table designed as instructed (3 points per field name)                      | 12                     |                      |
| PhoneNumber has an input mask (xxx) xxx-xxxx                                | 5                      |                      |
| Person records 1-5 are entered correctly (1 point per record)               | 5                      |                      |
| Person records 6-75 were imported correctly from National Student file      | 10                     |                      |
| Data Entry/Proofreading Errors -1 point each error (max -5 pts)             |                        |                      |
| <b>Subtotal 1a</b>                                                          | <b>37</b>              |                      |
| <b>Printout 1b – Relationship Report (35 points maximum)</b>                |                        |                      |
| One-to-Many relationships created (10 points each)                          | 20                     |                      |
| Referential integrity enforced (5 points each)                              | 10                     |                      |
| Printed with all three tables and relationships fully visible               | 5                      |                      |
| <b>Subtotal 1b</b>                                                          | <b>35</b>              |                      |
| <b>Printout 2a – Add Close Form Button (30 points maximum)</b>              |                        |                      |
| Command Button is added to the form                                         | 15                     |                      |
| Property Sheet, Event Tab shows On Click [event procedure]                  | 15                     |                      |
| <b>Subtotal 2a</b>                                                          | <b>30</b>              |                      |
| <b>Printout 2b – Conventions Form (85 points maximum)</b>                   |                        |                      |
| Designed as instructed (10 points per field)                                | 60                     |                      |
| Formatted correctly (should look like the example for print 2a)             | 15                     |                      |
| Printed record for all records                                              | 10                     |                      |
| Data Entry/Proofreading Errors -1 point each error (max -5 points)          |                        |                      |
| <b>Subtotal 2b</b>                                                          | <b>85</b>              |                      |
| <b>Printout 3a – Add Command Buttons (45 points maximum)</b>                |                        |                      |
| Three Command Buttons added to the form (10 points each)                    | 30                     |                      |
| Property Sheet, Event Tab shows On Click [event procedure] for Print Record | 15                     |                      |
| <b>Subtotal 3a</b>                                                          | <b>45</b>              |                      |
| <b>Printout 3b – Conventions Summary Form (60 points maximum)</b>           |                        |                      |
| Designed as instructed (5 points per field)                                 | 35                     |                      |
| Formatted correctly (should look like the example for print 3a)             | 15                     |                      |
| Printed only Beta Business Con record.                                      | 10                     |                      |
| Data Entry/Proofreading Errors -1 point each error (max -5 points)          |                        |                      |
| <b>Subtotal 3b</b>                                                          | <b>60</b>              |                      |

|                                                                                         |            |  |
|-----------------------------------------------------------------------------------------|------------|--|
| <b>Printout 4 – Convention Query (45 points maximum)</b>                                |            |  |
| Designed as instructed (5 points per field)                                             | 25         |  |
| Calculations for LeadTime are correct.                                                  | 10         |  |
| Show only records for ConventionID “5”                                                  | 10         |  |
| Data Entry/Proofreading Errors -1 point each error (max -5 points)                      |            |  |
| <b>Subtotal 4</b>                                                                       | <b>45</b>  |  |
| <b>Printout 5 – Convention Revenue Potential Query (60 points maximum)</b>              |            |  |
| Designed as instructed (5 points per field)                                             | 30         |  |
| Calculations for TotalRevenuePotential are correct                                      | 10         |  |
| Format results of TotalRevenuePotential as currency and no decimal places               | 10         |  |
| Show all records.                                                                       | 10         |  |
| Data Entry/Proofreading Errors -1 point each error (max -5 points)                      |            |  |
| <b>Subtotal 5</b>                                                                       | <b>60</b>  |  |
| <b>Printout 6 – Convention Revenue Report (52 points maximum)</b>                       |            |  |
| Title is Convention Revenue Report xx-xxxx-xxxx                                         | 5          |  |
| Correct fields are included in the report [including count and revenue] (2 points each) | 12         |  |
| Count formula for “Count” is correct                                                    | 10         |  |
| Count formula for “Revenue” is correct                                                  | 10         |  |
| Total Count formula for “Total Count” is correct                                        | 5          |  |
| Total Count formula for “Total Revenue” is correct                                      | 5          |  |
| Revenue results box is formatted for currency and no decimal points                     | 5          |  |
| Data Entry/Proofreading Errors -1 point each error (max -5 points)                      |            |  |
| <b>Subtotal 6</b>                                                                       | <b>52</b>  |  |
| <b>Total Points Earned</b>                                                              | <b>449</b> |  |

**Project:** NextGen Conference is a convention planning business that supports clients with the convention registration process. You have been tasked to develop a database to help make business decisions.

You need to create a database to keep track of all customers and their purchasing habits. Create a database named **National xx-xxxx-xxxx**, and save it as directed by the administrator.

**Job 1a: Conventions Table**

1. Create the table named **Conventions xx-xxxx-xxxx** shown below. Use the same field names and assign the best data type for each field. ConventionID is the primary key.
2. ConventionType should be a List Box with the following Value List:
  - a. business
  - b. education
  - c. k-12
  - d. sci-fi
  - e. technology
3. Enter all records shown below:

| ConventionID | ConventionName           | ConventionType | ConventionPrice | MaxAttendees | DateofConvention |
|--------------|--------------------------|----------------|-----------------|--------------|------------------|
| 1            | Alpha Conference         | sci-fi         | \$250           | 1200         | 9/28/2019        |
| 2            | Beta Business Con        | business       | 600             | 800          | 8/12/2019        |
| 3            | Charlie Annual Symposium | education      | 450             | 1500         | 8/01/2019        |
| 4            | Delta Tech Con           | technology     | 600             | 1800         | 10/01/2019       |
| 5            | Echo Ed Con              | k-12           | 525             | 500          | 9/15/2019        |

**Job 1b: Person Table (Printout #1a)**

1. Create a table named **Person xx-xxxx-xxxx** shown below. Use the same field names and assign the best data type for each field. PersonID is the primary key.
2. Enter all records shown below:

| PersonID | LastName   | FirstName | PhoneNumber    |
|----------|------------|-----------|----------------|
| 1        | Atwater    | Anika     | (824) 106-4234 |
| 2        | Jamison    | Jamika    | (597) 602-7108 |
| 3        | Koerber    | Kittie    | (742) 784-3556 |
| 4        | Alaniz     | Anjelica  | (200) 703-3986 |
| 5        | Millington | Michal    | (248) 464-4101 |

3. Import External Data to complete the addition of data for the table.
  - a. Choose New Data Source, File, Excel
  - b. Find the National\_Student File
  - c. Append a copy of the records to Person xx-xxxx-xxxx
4. Print the whole table in portrait orientation (records 1-75), two pages.

**Job 1c: Registration Table**

1. Create a table named Registration xx-xxxx-xxxx shown below. Use the same field names and assign the best data type for each field. EnrollmentID is the primary key.
2. Use the same name/data type for PersonID and ConventionID from their original tables.
  - d. Create Lookup List Box based on the Person Table (for PersonID field):
    - i. Bound Column = 1
    - ii. Columns = 3
    - iii. Column Heads = No
  - e. Create Lookup List Box based on the Convention Table (for ConventionID field):
    - i. Bound Column = 1
    - ii. Columns = 2
    - iii. Column Heads = No
3. Enter all records shown below:

| EnrollmentID | PersonID | ConventionID | DateEnrolled |
|--------------|----------|--------------|--------------|
| 1            | 11       | 3            | 3/14/2019    |
| 2            | 61       | 1            | 3/14/2019    |
| 3            | 45       | 2            | 3/14/2019    |
| 4            | 33       | 1            | 3/14/2019    |
| 5            | 53       | 4            | 3/14/2019    |
| 6            | 10       | 5            | 3/14/2019    |
| 7            | 13       | 4            | 3/14/2019    |
| 8            | 29       | 1            | 3/14/2019    |
| 9            | 14       | 3            | 3/15/2019    |
| 10           | 50       | 2            | 3/15/2019    |
| 11           | 13       | 1            | 3/15/2019    |
| 12           | 22       | 4            | 3/15/2019    |
| 13           | 5        | 5            | 3/15/2019    |
| 14           | 53       | 3            | 3/16/2019    |
| 15           | 1        | 5            | 3/31/2019    |
| 16           | 42       | 2            | 4/03/2019    |
| 17           | 37       | 2            | 4/03/2019    |
| 18           | 9        | 1            | 4/10/2019    |
| 19           | 42       | 4            | 4/20/2019    |
| 20           | 31       | 5            | 5/01/2019    |
| 21           | 61       | 5            | 5/03/2019    |
| 22           | 58       | 4            | 5/10/2019    |
| 23           | 63       | 2            | 5/22/2019    |
| 24           | 52       | 1            | 5/30/2019    |
| 25           | 28       | 5            | 5/31/2019    |

4. Create a Permanent Relationship (**Printout #1b**)
  - a. Create a permanent relationship between ConventionID in the Convention and Registration Tables.
    - a. Join ConventionID to ConventionID.
      - i. The Join Type should include ALL records from 'Convention xx-xxxx-xxxx' and only those records from 'Registrtration xx-xxxx-xxxx' where the joined fields are equal.
    - b. Create a permanent relationship between PersonID in the Person and Registration Tables.
      - i. The Join Type should include ALL records from 'Person xx-xxxx-xxxx' and only those records from 'Registration xx-xxxx-xxxx' where the joined fields are equal.
    - c. Enforce referential integrity.
    - d. Resize/move tables so each table name and each field is fully visible in the relationship window.
    - e. Create and **Print** the Relationship Report.

**Job 2: Conventions Form (Printouts #2a and #2b)**

1. Create a columnar form named Conventions Form xx-xxxx-xxxx using the following fields:
  - a. Convention Table: ConventionID, ConventionName, ConventionType, MaxAttendees, DateofConvention and ConventionPrice.
2. Change the ConventionType results box to combo box.
3. Add a command button to the form to Close the Form.
4. In Layout View, click on the button and display the property sheet, event tab.
5. Print Screen, paste in a new Word document and **PRINT** in landscape orientation.

Record: 1 of 5

Print 2a

6. **Print** the form for all the records in portrait orientation (two pages).

### Job 3: Conventions Summary Form (Printouts #3a and #3b)

1. Create a form with subform named Conventions Form xx-xxxx-xxxx (use the default name for the subform) using the following fields:
  - a. Convention Table: ConventionName, ConventionType, DateofConvention
  - b. Registration Table: PersonID, ConventionID, EnrollmentID, DateEnrolled
2. The form should be set as datasheet.
3. Change the Convention Type results box to combo box.
4. Add a three command button to the form (see example for layout)
  - a. Command button to Find a Record
  - b. Command button to Print a Record
  - c. Command button to Close the Form
5. In Layout View, click on the Print Record button and display the property sheet, event tab.
6. Print Screen, paste in a new word document and **PRINT** in landscape orientation.

The screenshot shows the Microsoft Access interface. The main window displays the 'Conventions Summary Form xx-xxxx-xxxx' in Design View. The form includes a subform 'Registration xx-xxxx-xxxx' which is set to 'Datasheet' view. The subform displays a table with the following data:

| PersonID | ConventionID | EnrollmentID | DateEnrolled |
|----------|--------------|--------------|--------------|
| 61       | 1            | 2            | 3/14/2019    |
| 33       | 1            | 4            | 3/14/2019    |
| 29       | 1            | 8            | 3/14/2019    |
| 13       | 1            | 11           | 3/15/2019    |
| 9        | 1            | 18           | 4/10/2019    |
| 52       | 1            | 24           | 5/30/2019    |
| *        | 1            | (New)        |              |

The main form has fields for 'ConventionName' (Alpha Conference), 'ConventionType' (sci-fi), and 'DateofConvention' (9/28/2019). At the bottom of the form, there are three command buttons: 'Find Record', 'Print Record' (highlighted with a red box), and 'Close Form'. The 'Property Sheet' is open on the right, showing the 'Print Record' button's properties, including 'On Click' and 'On Got Focus' events.

Print 3a

7. Use the Find Record button to find the “Beta Business Con” record.
8. Use the Print Record button to **PRINT** just the Beta Business Con record.

### Job 4: Convention Query (Printout #4)

1. Create a query named Convention Query xx-xxxx-xxxx using the tables/fields listed below:
  - a. Conventions Table: ConventionID, ConventionName
  - b. RegistrationTable: EnrollmentID, PersonID
2. Create a field called LeadTime. Calculate LeadTime by subtracting the DateEnrolled from the DateofConvention (results will be in days).
3. Restrict the results to only those by convention id “5”.
4. Print the results (showing ALL fields for convention id “5”) in landscape orientation. All data should be visible on one page.

**Job 5: Convention Revenue Potential Query (Printout #5)**

1. Create a query named Convention Revenue Potential Query xx-xxxx-xxxx using the tables/fields listed below:
  - a. Conventions Table: ConventionID, ConventionName, ConventionPrice, MaxAttendees
2. Create a field called TotalRevenuePotential. Calculate TotalRevenuePotential by multiplying the ConventionPrice by MaxAttendees.
3. Format the results of TotalRevenuePotential as currency and no decimal places.
4. Print the results (showing ALL fields) in landscape orientation. All data should be visible on one page.

**Job 6: Convention Revenue Report (Printout #6)**

1. Create a stepped report (landscape orientation), grouped by Convention, based on the following fields:
  - a. Conventions Table: ConventionName and ConventionPrice
  - b. Person Table: FirstName and LastName
2. Your report format, such as borders, colors, and placement of fields may be different, but it must:
  - a. Have the same title.
  - b. Show the same fields – resize and move to match the report layout shown.
  - c. In the ConventionName Footer:
    - i. Add a Text Box.
      1. Change the label to read “Count”
      2. Use the Count function to count the LastName field for the convention.
    - ii. Add a second Text Box.
      1. Change the label to read “Revenue”
      2. Use the Count function to calculate the number of LastName records multiplied by the ConventionPrice field
  - d. In the Report Footer
    - i. Add the same text boxes from the ConventionName Footer.
    - ii. Change the labels as follows:
      1. “Count” to “Total Count”
      2. “Revenue” to “Total Revenue”
  - e. Format the Revenue and Total Revenue result boxes as currency and no decimal places.
  - f. Have field widths/placement adjusted so all data is visible on one page in Landscape orientation.
  - g. **PRINT** the report (two pages).

Convention Revenue xx-xxxx-xxxx

| Convention Name   | Convention Price | First Name | Last Name |         |
|-------------------|------------------|------------|-----------|---------|
| Alpha Conference  | \$250.00         | Donald     | Dierks    |         |
|                   |                  | Mirtha     | Pipkins   |         |
|                   |                  | Crista     | Coria     |         |
|                   |                  | Shaun      | Volpe     |         |
|                   |                  | Leland     | Valentino |         |
|                   |                  | Howard     | Weis      |         |
| Count             |                  | 6          | Revenue   | \$1,500 |
| Beta Business Con | \$600.00         | Carlene    | Reynoso   |         |
|                   |                  | Dia        | McLeod    |         |
|                   |                  | Rosanna    | Hanline   |         |
|                   |                  | Missy      | Jump      |         |
|                   |                  | Delilah    | Kinsella  |         |
|                   |                  |            |           |         |